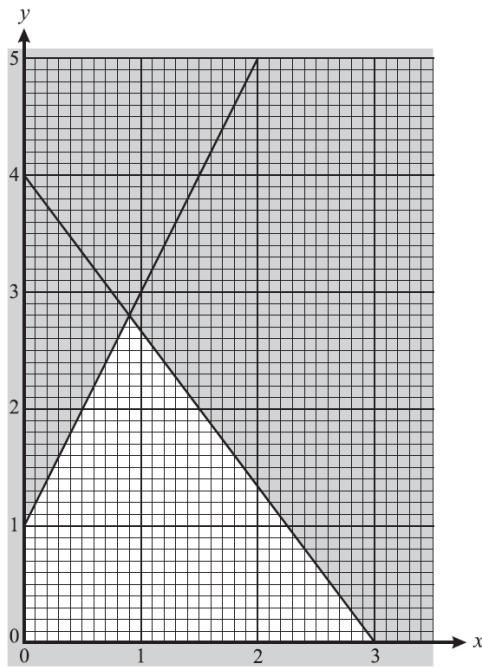


Discrete Revision: Linear programming

- 1 The constraints of a linear programming problem are represented by the graph below. The feasible region is the unshaded region, including its boundaries.



- (a) Write down four inequalities that define the feasible region.

Commented [NS(1): $x \geq 0$
 $y \geq 0$
 $4x + 3y \leq 12$
 $y \leq 2x + 1$

The objective is to maximise $P = 5x + 3y$.

- (b) Using the graph or otherwise, find the maximum value of P and the corresponding values of x and y .

Commented [NS(2): You can either put a ruler parallel to $5x + 3y = \text{constant}$ (draw this to show your method) or just evaluate P at the vertices. Because P increases with both x and y only the vertices $(3,0)$ where $P=15$ or $(0.9, 2.8)$ where $P=12.9$ could correspond to maximum values of P . So the maximum P is 15 at $(3,0)$

The objective is changed to maximise $Q = ax + 3y$.

- (c) For what set of values of a is the maximum value of Q equal to 3.

Commented [NS(3): Could Q equal 3 at $(3,0)$? This requires $a=1$, but then $Q=9.3$ at $(0.9, 2.8)$, so 3 won't be the maximum value of Q in this case.

Could Q equal 3 at $(0.9, 2.8)$? This requires $a=-6$ and in this case 3 is the maximum value of Q (anywhere on the line $y=2x+1$).

Could Q equal 3 at $(0,1)$? Well Q is always equal to 3 at $(0,1)$, so provided this point corresponds to the maximum value of Q we are fine. This point will give a maximum Q if $a \leq -6$.

Our final answer is $a \leq -6$

- 2 Dane is making the bridesmaid dresses and pageboy outfits for his sister's wedding. He expects it to take 20 hours to make each bridesmaid dress, and 15 hours to make each pageboy outfit. Each bridesmaid dress uses 8 metres of fabric, and each pageboy outfit uses 3 metres of fabric. The fabric costs £8 per metre. Additional items cost £36 for each bridesmaid dress and £96 for each pageboy outfit.

Dane's sister wants to have at least one bridesmaid and at least one pageboy. Dane has 100 hours of time available and must not spend more than £600 in total on materials.

Let x denote the number of bridesmaids and y denote the number of pageboys.

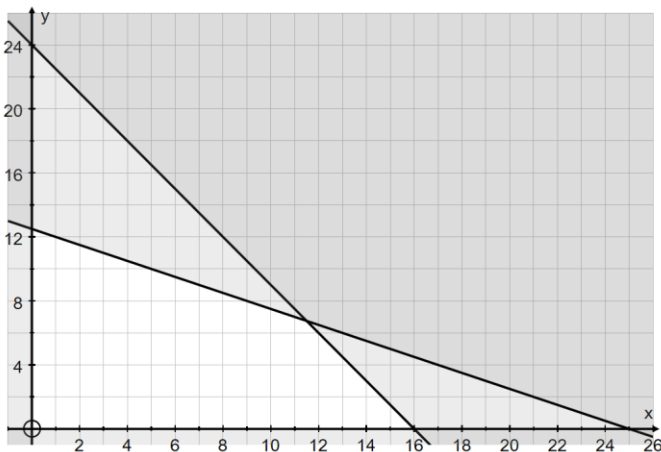
- (a) Show why the constraint $4x + 3y \leq 20$ is needed, and write down three more constraints on the values of x and y , other than that they must be integers.
- (b) Plot the feasible region where all four constraints are satisfied.

Dane's sister wants to maximum the total number of attendants (bridesmaids and pageboys) at her wedding.

- (c) Use your graph to find the maximum number of attendants.
- (d) What is the maximum amount of free time that Dane could have? What is the maximum amount of money that could be left over?

- 3 A company manufactures two types of potting compost, Flowerbase and Growmuch. The weekly amounts produced of each are constrained by the supplies of fibre and of nutrient mix. Each litre of Flowerbase requires 0.75 litres of fibre and 1 kg of nutrient mix. Each litre of Growmuch requires 0.5 litres of fibre and 2 kg of nutrient mix.

Let x and y be the weekly volumes, in thousands of litres, of Flowerbase and Growmuch produced respectively. The feasible values of x and y are shown in the graph below.

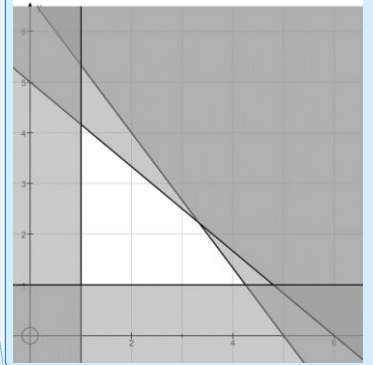


Commented [NS(4): The time to make x dresses and y outfits is $20x + 15y$. Because there are only 100 hours we must have $20x + 15y \leq 100$, which yields the given inequality once divided by 5.

We must have $x \geq 1$ and $y \geq 1$

Each dress costs $8 \times 8 + 36 = 100$ and each outfit costs $3 \times 8 + 96 = 120$. So $100x + 120y \leq 600$

Commented [NS(5):



Commented [NS(6): We need to maximise $P = x + y$, and only integer points are allowed. The points (1,4), (2,3), (3,2), (4,1) all give five attendants in total. It is clearly not possible to have 6, because the line $x + y = 6$ doesn't intersect the feasible region.

Commented [NS(7): If the point (1,4) is used, then the total time is 80 and the total cost is 580. So there is 20 hours and £20 left over.

If the point (2,3) is used then the total time is 85 and the total cost is 560. So there are 15 hours and £40 left over.

If the point (3,2) is used then the total time is 90 and the total cost is 540. So there are 10 hours and £60 left over.

If the point (4,1) is used then the total time is 95 and the total cost is 520. So there are 5 hours, and £80 left over.

So the maximum amount of free time is 20 hours, and the maximum amount of money left over is £80.

- (a) Find the weekly amounts of fibre and nutrient mix available to the company.

The profit on Flowerbase is 9p per litre. The profit on Growmuch is 20p per litre.

- (b) Find the maximum weekly profit the company can make. Give the weekly volumes of Flowerbase and Growmuch that the company should produce.
- (c) Consider each of the following separate circumstances:

- (i) There is a reduction in the weekly supply of fibre to 10,000 litres. What effect does this have on profit?
- (ii) The price of fibre is increased. Will this affect the optimal production plan? Justify your answer.
- (iii) The supply of nutrient mix is increased to 30,000 kg per week. What is the new profit?

Commented [NS(8): You need to identify which line is which, or at least recognise that Flowerbase needs more fibre, so the fibre available is $16 \times 0.75 = 12,000$ litres, and the nutrient mix is 25 tonnes. Note the units here because x and y are in thousands of litres.

Commented [NS(9): Don't just assume that the maximum value will occur at the point of intersection. You need to either draw on some lines of constant profit, or calculate the profit at each of the vertices of the feasible region (except the origin).

At (16,0) $P=144$, at (0,12.5) $P=250$ and at (11.5, 6.75) $P=238.5$.

So, taking into account units, the maximum profit is £2500 per week, and this is obtained by manufacturing no Flowerbase, and 12,500 litres of Growmuch.

Commented [NS(10): Lowering the fibre line (by a bit) will have no effect on the vertex corresponding to maximum profit.

Commented [NS(11): Because Flowerbase uses more fibre than Growmuch, the profit per litre of Flowerbase will decrease by more than the profit per litre of Growmuch. Therefore the optimal production plan will be unaffected.

Commented [NS(12): The point of intersection with the y -axis increases to (15,0), corresponding to a profit of £3000.

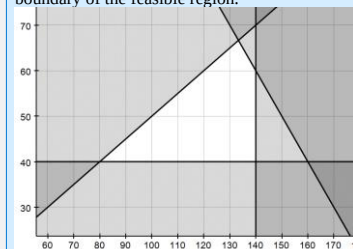
Because the gradients of the lines haven't changed, the maximum profit must still correspond to this point. You could also check this by finding the new point of intersection between fibre and nutrient and verifying that the profit (£2910) is less than £3000.

Commented [NS(13): Minimise $P=8x+10y+14z$ subject to
 $x+y+z \geq 200$
 $2x \geq 5y$
 $3x \leq 7y+7z$
 $4y \geq x+z$
 $x \geq 0, y \geq 0, z \geq 0$

Commented [NS(14): If you write $z=200-x-y$ and substitute into your answer from Part (a) then the problem becomes maximise $P=3x+2y$ subject to

$2x \geq 5y$
 $x \leq 140$
 $y \geq 40$
 $x+y \leq 200$

Note that the last of these constraints comes from the requirement that $z \geq 0$, and forms a crucial part of the boundary of the feasible region.



Evaluate P at the points of intersection, or consider the gradients of the objective line, and you should decide that P is maximised at (140,60). So they should make 140 ring, 60 jam, and no custard doughnuts.

- 4 A bakery makes three types of doughnut: ring, jam, and custard. The bakery has the following constraints on the number of doughnuts it must make each day:

- The total number of doughnuts made must be at least 200.
- They must make at least twice as many ring doughnuts as jam doughnuts.
- At most 70% of the doughnuts the bakery makes must be ring doughnuts.
- At least a fifth of the doughnuts the bakery makes must be jam doughnuts.

It costs 8 pence to make each ring doughnut, 10 pence to make each jam doughnut, and 14 pence to make each custard doughnut. The bakery wants to minimise the total daily costs of making the required doughnuts.

Let x represent the number of ring doughnuts, let y represent the number of jam doughnuts, and let z represent the number of custard doughnuts the bakery makes each day.

- (a) Formulae this as a linear programming problem, stating the objective and listing the constraints as simplified inequalities with integer coefficients.

One day, instead of making at least 200 doughnuts, the bakery requires that **exactly 200** doughnuts are made.

- (b) Determine the number each type of doughnut the bakery should make on that day.

- 5 The table shows the butter and sugar content in two recipes. The first recipe is for 1 kg of toffee and the second is for 1 kg of fudge.

	Toffee	Fudge
Butter	100 g	150 g
Sugar	800 g	700 g

A confectioner has 1.5 kg of butter, and 10 kg of sugar available. There are no constraints on the availability of other ingredients.

- (a) What is the maximum amount of toffee which the confectioner could make? How much butter or sugar would be left over?

What is the maximum amount of fudge which the confectioner could make? How much butter or sugar would be left over?

- (b) Formulate a linear programming problem to find the maximum total amount of toffee and fudge which the confectioner can make.

Solve your linear programming problem graphically.

The confectioner charges £5.50 for 1 kg of toffee and £4.50 for 1 kg of fudge.

- (c) What quantities should she make to maximise her income? Justify your answer.
- (d) By how much would the price of toffee have to change for the maximum income solution to change?

- 6 The graph below shows the feasible region in a linear programming problem.



The constraints are $x + y \leq 10$, $4x + y \leq 28$, and $2x + 5y \leq 41$.

The objective is to maximise the value of $P = kx + 3y$, where $k \geq 0$.

Find the range of values of k for which (3,7) is an optimal point.

Commented [NS(15): There is butter for 15kg of toffee and sugar for 12.5 kg of toffee. Therefore, the maximum amount of toffee is 12.5 kg, and there will be 250 g of butter left over.

Commented [NS(16): There is enough butter for 10 kg of fudge, and enough sugar for 14.3 kg of fudge. So the maximum amount of fudge is 10 kg, and this will leave 3 kg of sugar left over.

Commented [NS(17): Maximise $P=x+y$ subject to
 $2x+3y \leq 30$
 $8x+7y \leq 100$
 $x \geq 0$
 $y \geq 0$

Do include the last two inequalities, even though they might seem obvious.

Commented [NS(18): If you take the effort to plot the feasible region then you will be able to see that the vertices of the feasible region are (12.5, 0), (9,4), and (0,10). You could also do the same with a sketch, but you'll have to calculate the point of intersection between the butter and sugar lines.

The maximum amount of product is clearly 13, when 9 kg of toffee, and 4 kg of fudge are made. You could also obtain this from a plot by sliding your ruler parallel to a line of constant profit. Do draw at least one of these lines to make your method clear.

Commented [NS(19): If you plotted a graph, then you can slide a ruler parallel to $5.5x+4.5y=\text{constant}$. If you sketched, you'll have to do the calculations: $P=£68.75$ at (12.5,0), $P=£67.5$ at (9,4) and $P=£45$ at (0,10). So the confectioner should make 12.5 of toffee and no fudge.

Commented [NS(20): To answer this you can either consider gradients, or just do the calculations at the other points. Let's do the latter:

In order to make more profit at the point (9,4) we need $12.5t < 9t + 4 \times 4.5$ so $t < 5.143$. Because money is to the nearest penny, this means that if the price of toffee falls to £5.14 or below, then the maximum income solution will change.

Commented [NS(21): $1.2 \leq k \leq 3$.

Using the vertices we need $3k+21 > 24.6$ and $3k+21 = 6k+12$.

Using gradients we need $k/3 > 2/5$ and $k/3 \leq 1$